

Lab 3 – Architectural Pattern Selection

Dropshipping Inventory & Order Orchestrator | Microservices Architecture

1. Architecture Selection

We chose **Microservices Architecture for the Dropshipping Inventory & Order Orchestrator**. The architecture separates major responsibilities into independently deployable services: Order Manager, Payment Service, Supplier Inventory Service, Supplier Purchase Order Service, and Shipment Tracking Service.

2. Reason 1 – Independent Scaling

The system performs different operations such as order processing, supplier inventory verification, purchase-order dispatch, and shipment tracking. These services can be scaled independently according to their workload instead of scaling the complete application.

3. Reason 2 – Fault Isolation and Independent Deployment

Separating major functions reduces the impact of failures in individual services. For example, a temporary problem in shipment tracking does not have to make the complete order-processing functionality unavailable. Individual services can also be updated and deployed independently.

4. Security Advantage

The microservices structure allows sensitive operations such as payment processing and supplier communication to be isolated behind dedicated service interfaces. Access between services can be controlled through authenticated and authorized API communication, limiting unnecessary exposure of internal functionality.

5. Performance Benefit

Independent services can be scaled according to demand. Frequently used operations such as order processing and supplier inventory verification can receive additional resources without scaling unrelated services. This can improve responsiveness when multiple orders are processed concurrently.

6. Components Used

Component	Primary Responsibility
Order Manager	Coordinates order processing and service requests.
Payment Service	Handles payment-processing requests.
Supplier Inventory Service	Verifies supplier SKU inventory availability.
Supplier Purchase Order Service	Creates and dispatches supplier purchase orders.
Shipment Tracking Service	Handles shipment/tracking information and status updates.

This justification follows the Lab 3 requirement to state the architectural choice, give two scenario-specific reasons, and explain one security advantage and one performance benefit.